

Binomial Theorem related Proofs — Solutions

Warm-up

Question 1:

We use the definition of the factorial $n! = n \times (n-1) \times (n-2) \times \dots \times 1$.

$$\begin{aligned}\frac{(n+1)!}{n!} &= \frac{(n+1) \times n \times (n-1) \times \dots \times 1}{n \times (n-1) \times \dots \times 1} \\ &= n+1\end{aligned}$$

Question 2:

The general term of the binomial expansion $(1+x)^n$ is given by $T_{r+1} = \binom{n}{r}x^r$

$$\begin{aligned}(1+x)^n &= \sum_{r=0}^n \binom{n}{r}x^r \\ &= \binom{n}{0}x^0 + \binom{n}{1}x^1 + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots\end{aligned}$$

Question 3:

Using the factorial definition $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

$$\begin{aligned}\binom{n}{n-r} &= \frac{n!}{(n-r)!(n-(n-r))!} \\ &= \frac{n!}{(n-r)!r!} \\ &= \frac{n!}{r!(n-r)!} \\ &= \binom{n}{r}\end{aligned}$$

Question 4:

Starting with the left-hand side (LHS):

$$\begin{aligned}
LHS &= r \binom{n}{r} \\
&= r \times \frac{n!}{r!(n-r)!} \\
&= r \times \frac{n!}{r \times (r-1)!(n-r)!} \\
&= \frac{n!}{(r-1)!(n-r)!} \\
&= n \times \frac{(n-1)!}{(r-1)!(n-r)!} \\
&= n \times \frac{(n-1)!}{(r-1)!((n-1)-(r-1))!} \\
&= n \binom{n-1}{r-1}
\end{aligned}$$

Question 5:

The sum of coefficients is found by setting the variable x to 1.

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n$$

Substitute $x = 1$:

$$\begin{aligned}
(1+1)^n &= \binom{n}{0} + \binom{n}{1}(1) + \binom{n}{2}(1)^2 + \dots + \binom{n}{n}(1)^n \\
2^n &= \sum_{r=0}^n \binom{n}{r}
\end{aligned}$$

Question 6:

We substitute $x = -1$ into the binomial expansion of $(1+x)^5$.

$$(1+x)^5 = \binom{5}{0} + \binom{5}{1}x + \binom{5}{2}x^2 + \binom{5}{3}x^3 + \binom{5}{4}x^4 + \binom{5}{5}x^5$$

Substitute $x = -1$:

$$\begin{aligned}
(1-1)^5 &= \binom{5}{0} + \binom{5}{1}(-1) + \binom{5}{2}(-1)^2 + \binom{5}{3}(-1)^3 + \binom{5}{4}(-1)^4 + \binom{5}{5}(-1)^5 \\
0 &= \binom{5}{0} - \binom{5}{1} + \binom{5}{2} - \binom{5}{3} + \binom{5}{4} - \binom{5}{5}
\end{aligned}$$

Question 7:

The given sum is $\sum_{r=0}^{10} \binom{10}{r} 2^r$. This matches the form of the expansion $(1+x)^n = \sum_{r=0}^n \binom{n}{r} x^r$ where $n = 10$ and $x = 2$.

$$(1+x)^{10} = \sum_{r=0}^{10} \binom{10}{r} x^r$$

Substitute $x = 2$:

$$(1+2)^{10} = \sum_{r=0}^{10} \binom{10}{r} 2^r$$

$$3^{10} = \sum_{r=0}^{10} \binom{10}{r} 2^r$$

Question 8:

The sum $\sum_{k=0}^4 \binom{4}{k} 10^k$ is the expansion of $(1+x)^n$ with $n = 4$ and $x = 10$.

$$\begin{aligned} \sum_{k=0}^4 \binom{4}{k} 10^k &= (1+10)^4 \\ &= 11^4 \\ &= 14641 \end{aligned}$$

Question 9:

This is the expansion of $(1+x)^8$ where $x = -1$.

$$(1+x)^8 = \sum_{r=0}^8 \binom{8}{r} x^r$$

Substitute $x = -1$:

$$\begin{aligned} (1-1)^8 &= \binom{8}{0} + \binom{8}{1}(-1) + \binom{8}{2}(-1)^2 + \dots + \binom{8}{8}(-1)^8 \\ 0 &= \binom{8}{0} - \binom{8}{1} + \binom{8}{2} - \dots + \binom{8}{8} \end{aligned}$$

Question 10:

The sum of coefficients is found by setting $x = 1$ in the expansion $(2+x)^n = \sum_{r=0}^n \binom{n}{r} 2^{n-r} x^r$.

$$\begin{aligned} \text{Sum of coefficients} &= (2+1)^n \\ &= 3^n \end{aligned}$$

Question 11:

The series is $1 + 3\binom{n}{1} + 9\binom{n}{2} + \dots + 3^n\binom{n}{n}$.

$$\begin{aligned}
\text{Series} &= \binom{n}{0}3^0 + \binom{n}{1}3^1 + \binom{n}{2}3^2 + \dots + \binom{n}{n}3^n \\
&= \sum_{r=0}^n \binom{n}{r}3^r \\
&= (1+3)^n \\
&= 4^n
\end{aligned}$$

Thus, the binomial is $(1+x)^n$ and the value of x is 3.

Skill-building

Question 1:

Consider the expansion of $(1+x)^n = \sum_{r=0}^n \binom{n}{r}x^r$.

Substitute $x = -1$:

$$\begin{aligned}
(1-1)^n &= \sum_{r=0}^n \binom{n}{r}(-1)^r \\
0 &= \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \dots + (-1)^n \binom{n}{n}
\end{aligned}$$

Question 2:

We are looking for the coefficient of x^2 in $(1+x)^5(1+x)^2 = (1+x)^7$.

$$\begin{aligned}
(1+x)^7 &= \sum_{r=0}^7 \binom{7}{r}x^r \\
\text{Coefficient of } x^2 &= \binom{7}{2} \\
&= \frac{7 \times 6}{2 \times 1} \\
&= 21
\end{aligned}$$

Question 3:

Using the factorial definition of the binomial coefficient $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

$$\begin{aligned}
\frac{1}{r+1} \binom{n}{r} &= \frac{1}{r+1} \times \frac{n!}{r!(n-r)!} \\
&= \frac{n!}{(r+1)r!(n-r)!} \\
&= \frac{n!}{(r+1)!(n-r)!} \\
\text{Multiply by } \frac{n+1}{n+1} &= \frac{(n+1)n!}{(n+1)(r+1)!(n-r)!} \\
&= \frac{(n+1)!}{(n+1)(r+1)!(n-r)!} \\
&= \frac{1}{n+1} \times \frac{(n+1)!}{(r+1)!((n+1)-(r+1))!} \\
&= \frac{1}{n+1} \binom{n+1}{r+1}
\end{aligned}$$

Question 4:

Consider the expansion of $(1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$.

Substitute $x = 3$:

$$\begin{aligned}
(1+3)^n &= \sum_{k=0}^n \binom{n}{k} 3^k \\
4^n &= \sum_{k=0}^n 3^k \binom{n}{k}
\end{aligned}$$

Question 5:

Using the factorial definition $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

$$\begin{aligned}
LHS &= \binom{n}{r} \times \binom{r}{k} \\
&= \frac{n!}{r!(n-r)!} \times \frac{r!}{k!(r-k)!} \\
&= \frac{n!}{(n-r)!k!(r-k)!} \\
\text{Rearrange terms} &= \frac{n!}{k!(n-k)!} \times \frac{(n-k)!}{(n-r)!(r-k)!} \\
&= \binom{n}{k} \times \binom{n-k}{r-k}
\end{aligned}$$

Question 6:

Using the factorial formula $\binom{n}{r} = \frac{n!}{r!(n-r)!}$

$$\begin{aligned}
LHS &= \binom{2n}{2} \\
&= \frac{(2n)!}{2!(2n-2)!} \\
&= \frac{(2n)(2n-1)(2n-2)!}{2(2n-2)!} \\
&= n(2n-1) \\
&= 2n^2 - n
\end{aligned}$$

Now for the RHS:

$$\begin{aligned}
RHS &= 2\binom{n}{2} + n^2 \\
&= 2 \times \frac{n!}{2!(n-2)!} + n^2 \\
&= 2 \times \frac{n(n-1)(n-2)!}{2(n-2)!} + n^2 \\
&= n(n-1) + n^2 \\
&= n^2 - n + n^2 \\
&= 2n^2 - n
\end{aligned}$$

Since $LHS = RHS$, the identity is proven.

Question 7:

The product is $(1+x)^n(1+x)^n = (1+x)^{2n}$.

$$\text{Coefficient of } x^n \text{ in } (1+x)^{2n} = \binom{2n}{n}$$

Now, consider the product of two expansions:

$$(1+x)^n(1+x)^n = \left(\sum_{r=0}^n \binom{n}{r} x^r \right) \times \sum_{k=0}^n \binom{n}{k} x^k$$

To get x^n , we need $r+k=n$, which means $k=n-r$.

$$\text{Coefficient of } x^n = \sum_{r=0}^n \binom{n}{r} \binom{n}{n-r}$$

$$\text{Since } \binom{n}{n-r} = \binom{n}{r} \text{ (from Warm-up 3)}$$

$$\begin{aligned}
\text{Coefficient of } x^n &= \sum_{r=0}^n \binom{n}{r} \binom{n}{r} \\
&= \sum_{r=0}^n \binom{n}{r}^2
\end{aligned}$$

Question 8:

We expand $(1 - x^2)^n$ in two ways.

$$\begin{aligned}\text{Method 1: } (1 - x^2)^n &= \sum_{k=0}^n \binom{n}{k} (-x^2)^k \\ &= \sum_{k=0}^n \binom{n}{k} (-1)^k x^{2k}\end{aligned}$$

$$\text{Method 2: } (1 - x)^n (1 + x)^n = \left(\sum_{i=0}^n \binom{n}{i} (-x)^i \right) \times \sum_{j=0}^n \binom{n}{j} x^j$$

To find the coefficient of x^n in Method 2, we need $i + j = n$, so $j = n - i$.

$$\begin{aligned}\text{Coefficient of } x^n &= \sum_{i=0}^n \binom{n}{i} (-1)^i \binom{n}{n-i} \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i} \binom{n}{i} \\ &= \sum_{i=0}^n (-1)^i \binom{n}{i}^2\end{aligned}$$

Equating the coefficients of x^n from both methods:

$$\begin{aligned}\text{If } n \text{ is odd, } 2k = n \text{ has no integer solution for } k. \\ \implies \text{Coefficient of } x^n \text{ in } (1 - x^2)^n &= 0 \\ \implies \sum_{k=0}^n (-1)^k \binom{n}{k}^2 &= 0\end{aligned}$$

Question 9:

Using Pascal's Identity $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$.

$$\begin{aligned}\binom{n+2}{r} &= \binom{n+1}{r} + \binom{n+1}{r-1} \\ &= \left(\binom{n}{r} + \binom{n}{r-1} \right) + \left(\binom{n+1}{r-1} \right)\end{aligned}$$

$$\text{Apply Pascal's again to } \binom{n+1}{r-1} = \binom{n}{r-1} + \binom{n}{r-2}$$

$$\begin{aligned}\text{Substitute back} &= \binom{n}{r} + \binom{n}{r-1} + \binom{n}{r-1} + \binom{n}{r-2} \\ &= \binom{n}{r} + 2\binom{n}{r-1} + \binom{n}{r-2}\end{aligned}$$

Question 10:

Consider the identity $(1+x)^m(1+x)^n = (1+x)^{m+n}$.

$$\begin{aligned}\text{LHS: } (1+x)^m(1+x)^n &= \sum_{k=0}^m \binom{m}{k} x^k \times \sum_{j=0}^n \binom{n}{j} x^j \\ &= \sum_{k=0}^m \sum_{j=0}^n \binom{m}{k} \binom{n}{j} x^{k+j}\end{aligned}$$

To find the coefficient of x^r , we set $k+j=r$, so $j=r-k$.

$$\text{Coefficient of } x^r = \sum_{k=0}^r \binom{m}{k} \binom{n}{r-k}$$

$$\text{RHS: } (1+x)^{m+n} = \sum_{r=0}^{m+n} \binom{m+n}{r} x^r$$

Equating the coefficients of x^r on both sides:

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

Easier Exam Questions**Question 1:**

We use the factorial definition $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\begin{aligned}\binom{n+2}{3} - \binom{n}{3} &= \frac{(n+2)!}{3!(n-1)!} - \frac{n!}{3!(n-3)!} \\ &= \frac{(n+2)(n+1)n(n-1)!}{6(n-1)!} - \frac{n(n-1)(n-2)(n-3)!}{6(n-3)!} \\ &= \frac{n(n^2+3n+2)}{6} - \frac{n(n^2-3n+2)}{6} \\ &= \frac{n}{6}[(n^2+3n+2) - (n^2-3n+2)] \\ &= \frac{n}{6}[6n] \\ &= n^2\end{aligned}$$

Question 2:

Using the notation ${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\begin{aligned}
\frac{{}^{n+1}C_r}{{}^nC_{r-1}} &= \frac{\frac{(n+1)!}{r!(n+1-r)!}}{\frac{n!}{(r-1)!(n-(r-1))!}} \\
&= \frac{(n+1)!}{r!(n+1-r)!} \times \frac{(r-1)!(n-r+1)!}{n!} \\
&= \frac{(n+1)n!}{n!} \times \frac{(r-1)!}{r!} \times \frac{(n-r+1)!}{(n-r+1)!} \\
&= (n+1) \times \frac{1}{r} \times 1 \\
&= \frac{n+1}{r}
\end{aligned}$$

Question 3:

i) The binomial expansion of $(1+x)^n$ is:

$$\begin{aligned}
(1+x)^n &= \sum_{r=0}^n \binom{n}{r} x^r \\
&= \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n
\end{aligned}$$

ii) To show that ${}^nC_0 + {}^nC_1 + \dots + {}^nC_n = 2^n$, we substitute $x = 1$ into the expansion:

$$\begin{aligned}
(1+1)^n &= \binom{n}{0} + \binom{n}{1}(1) + \binom{n}{2}(1)^2 + \dots + \binom{n}{n}(1)^n \\
2^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n}
\end{aligned}$$

Question 4:

We use the identity $\sum_{r=0}^n \binom{n}{r} = 2^n$ derived in the previous question.

$$\begin{aligned}
\text{Sum} &= \binom{100}{0} + \binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100} \\
&= \sum_{r=0}^{100} \binom{100}{r} \\
&= 2^{100}
\end{aligned}$$

Question 5:

Consider the expansion of $(1+x)^{10} = \sum_{r=0}^{10} \binom{10}{r} x^r$.

$$(1+x)^{10} = \binom{10}{0} + \binom{10}{1}x + \binom{10}{2}x^2 + \dots + \binom{10}{10}x^{10}$$

Substitute $x = 5$:

$$(1+5)^{10} = \binom{10}{0} + \binom{10}{1}(5) + \binom{10}{2}(5)^2 + \dots + \binom{10}{10}(5)^{10}$$

$$6^{10} = 1 + [5\binom{10}{1} + 5^2\binom{10}{2} + \dots + 5^{10}\binom{10}{10}]$$

$$6^{10} - 1 = 5\binom{10}{1} + 5^2\binom{10}{2} + \dots + 5^{10}\binom{10}{10}$$

Thus, the correct option is (D).

Question 6:

We use the identity $(1+x)^4(1+x)^9 = (1+x)^{13}$.

$$\text{LHS: } (1+x)^4(1+x)^9 = \sum_{i=0}^4 \binom{4}{i}x^i \times \sum_{j=0}^9 \binom{9}{j}x^j$$

$$\begin{aligned} \text{Coefficient of } x^4 &= \sum_{i=0}^4 \binom{4}{i} \binom{9}{4-i} \\ &= \binom{4}{0}\binom{9}{4} + \binom{4}{1}\binom{9}{3} + \binom{4}{2}\binom{9}{2} + \binom{4}{3}\binom{9}{1} + \binom{4}{4}\binom{9}{0} \end{aligned}$$

$$\text{RHS: } (1+x)^{13} = \sum_{r=0}^{13} \binom{13}{r}x^r$$

$$\text{Coefficient of } x^4 = \binom{13}{4}$$

Equating the coefficients of x^4 on both sides proves the identity.

Question 7:

We use the identity $(1+x)^{10} = (1+x)^5(1+x)^5$.

$$\text{LHS: } (1+x)^{10} \text{ coeff of } x^3 = \binom{10}{3}$$

$$\text{RHS: } (1+x)^5(1+x)^5 = \sum_{i=0}^5 \binom{5}{i}x^i \times \sum_{j=0}^5 \binom{5}{j}x^j$$

$$\begin{aligned} \text{Coeff of } x^3 &= \sum_{i=0}^3 \binom{5}{i} \binom{5}{3-i} \\ &= \binom{5}{0}\binom{5}{3} + \binom{5}{1}\binom{5}{2} + \binom{5}{2}\binom{5}{1} + \binom{5}{3}\binom{5}{0} \end{aligned}$$

Equating the coefficients of x^3 proves the identity.

Question 8:

We use the identity $(1+x)^{11} = (1+x)^8(1+x)^3$.

$$\text{LHS: } (1+x)^{11} \text{ coeff of } x^3 = \binom{11}{3}$$

$$\text{RHS: } (1+x)^8(1+x)^3 = \sum_{i=0}^8 \binom{8}{i} x^i \times \sum_{j=0}^3 \binom{3}{j} x^j$$

$$\begin{aligned} \text{Coeff of } x^3 &= \sum_{i=0}^3 \binom{8}{i} \binom{3}{3-i} \\ &= \binom{8}{0} \binom{3}{3} + \binom{8}{1} \binom{3}{2} + \binom{8}{2} \binom{3}{1} + \binom{8}{3} \binom{3}{0} \end{aligned}$$

Equating the coefficients of x^3 proves the identity.

Harder Exam Questions

Question 1:

i) The binomial expansion of $(1+x)^n$ is:

$$\begin{aligned} (1+x)^n &= \sum_{r=0}^n \binom{n}{r} x^r \\ &= \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n \end{aligned}$$

ii) To show that $\binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} = 2^n - 1$, we substitute $x = 1$:

$$\begin{aligned} (1+1)^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \\ 2^n &= 1 + \left[\binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \right] \\ 2^n - 1 &= \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \dots + \binom{n}{n} \end{aligned}$$

iii) To show $1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1} = 3^n - 2^n$, we substitute $x = 2$ into $(1+x)^n$:

$$\begin{aligned}
(1+2)^n &= \binom{n}{0} + \binom{n}{1}(2) + \binom{n}{2}(2)^2 + \dots + \binom{n}{n}(2)^n \\
3^n &= 1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^n\binom{n}{n} \\
3^n &= 1 + [2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1}] + 2^n \\
3^n - 2^n &= 1 + 2\binom{n}{1} + 4\binom{n}{2} + \dots + 2^{n-1}\binom{n}{n-1}
\end{aligned}$$

Question 2:

b. We use the identity $(1+x)^{11} = (1+x)^3(1+x)^8$.

$$\text{LHS: } (1+x)^{11} \text{ coeff of } x^5 = \binom{11}{5}$$

$$\begin{aligned}
\text{RHS: } (1+x)^3(1+x)^8 &= \sum_{i=0}^3 \binom{3}{i} x^i \times \sum_{j=0}^8 \binom{8}{j} x^j \\
\text{Coeff of } x^5 &= \binom{3}{0}\binom{8}{5} + \binom{3}{1}\binom{8}{4} + \binom{3}{2}\binom{8}{3} + \binom{3}{3}\binom{8}{2} \\
&= 1\binom{8}{5} + 3\binom{8}{4} + 3\binom{8}{3} + 1\binom{8}{2}
\end{aligned}$$

Equating the coefficients of x^5 proves the identity: ${}^{11}C_5 = {}^8C_5 + {}^3C_1 {}^8C_4 + {}^3C_2 {}^8C_3 + {}^8C_2$.

e. We use the expansion of $(1+x)^{2n} = \sum_{r=0}^{2n} \binom{2n}{r} x^r$.

Substitute $x = 1$:

$$\begin{aligned}
(1+1)^{2n} &= \sum_{r=0}^{2n} \binom{2n}{r} \\
2^{2n} &= \binom{2n}{0} + \binom{2n}{1} + \dots + \binom{2n}{n-1} + \binom{2n}{n} + \binom{2n}{n+1} + \dots + \binom{2n}{2n}
\end{aligned}$$

$$\text{Using } \binom{2n}{r} = \binom{2n}{2n-r} \text{ (symmetry)}$$

$$2^{2n} = 2\left[\binom{2n}{0} + \binom{2n}{1} + \dots + \binom{2n}{n-1}\right] + \binom{2n}{n}$$

$$2^{2n} = 2\sum_{r=0}^{n-1} \binom{2n}{r} + \binom{2n}{n}$$

$$\text{Alternatively, } 2\sum_{r=0}^n \binom{2n}{r} = 2\left[\binom{2n}{0} + \dots + \binom{2n}{n}\right]$$

$$= 2\sum_{r=0}^{n-1} \binom{2n}{r} + 2\binom{2n}{n}$$

$$\Rightarrow 2\sum_{r=0}^n \binom{2n}{r} - \binom{2n}{n} = 2\sum_{r=0}^{n-1} \binom{2n}{r} + \binom{2n}{n} = 2^{2n}$$

f. (i) Consider the expansions of $(q + p)^n$ and $(q - p)^n$.

$$(q + p)^n = \sum_{r=0}^n \binom{n}{r} q^{n-r} p^r$$

$$(q - p)^n = \sum_{r=0}^n \binom{n}{r} q^{n-r} (-p)^r$$

$$(q + p)^n - (q - p)^n = \sum_{r=0}^n \binom{n}{r} q^{n-r} p^r - \sum_{r=0}^n \binom{n}{r} q^{n-r} (-1)^r p^r$$

$$= \sum_{r=0}^n \binom{n}{r} q^{n-r} p^r [1 - (-1)^r]$$

If r is even, $1 - (-1)^r = 0$.

If r is odd, $1 - (-1)^r = 2$.

$$\Rightarrow (q + p)^n - (q - p)^n = 2 \left[\binom{n}{1} q^{n-1} p^1 + \binom{n}{3} q^{n-3} p^3 + \dots \right]$$

(ii) If n is odd, the last term in the sum is $r = n$.

$$\text{Last term} = 2 \binom{n}{n} q^{n-n} p^n = 2p^n$$

(iii) If n is even, the last odd r is $n - 1$.

$$\text{Last term} = 2 \binom{n}{n-1} q^{n-(n-1)} p^{n-1} = 2 \binom{n}{n-1} q p^{n-1}$$

Question 3:

We substitute $x = 2$ into the expansion $(1 + x)^n = \sum_{r=0}^n \binom{n}{r} x^r$.

$$(1 + 2)^n = \binom{n}{0} + \binom{n}{1}(2) + \binom{n}{2}(2)^2 + \dots + \binom{n}{n}(2)^n$$

$$3^n = 1 + 2 \binom{n}{1} + 4 \binom{n}{2} + \dots + 2^n \binom{n}{n}$$

$$3^n - 2^n = 1 + 2 \binom{n}{1} + 4 \binom{n}{2} + \dots + 2^{n-1} \binom{n}{n-1}$$

Question 4:

i) Using the factorial definition $\binom{n}{r} = \frac{n!}{r!(n-r)!}$.

$$\begin{aligned}
{}^{n+1}C_r - {}^nC_r &= \binom{n+1}{r} - \binom{n}{r} \\
&= \frac{(n+1)!}{r!(n+1-r)!} - \frac{n!}{r!(n-r)!} \\
&= \frac{(n+1)n!}{r!(n+1-r)!} - \frac{n!(n+1-r)}{r!(n+1-r)!} \\
&= \frac{n!}{r!(n+1-r)!} [(n+1) - (n+1-r)] \\
&= \frac{n!}{r!(n+1-r)!} [r] \\
&= \frac{n!}{(r-1)!(n+1-r)!} \\
&= \binom{n}{r-1}
\end{aligned}$$

ii) We are asked to find the sum $S = \binom{3}{2} + \binom{4}{2} + \binom{5}{2} + \dots + \binom{100}{2}$.

From (i), $\binom{n}{r} = \binom{n+1}{r} - \binom{n+1}{r+1}$ is not quite right. Let's use $\binom{n}{r-1} = \binom{n+1}{r} - \binom{n}{r}$.

Wait, the identity is $\binom{n}{r-1} = \binom{n+1}{r} - \binom{n}{r}$. For $r = 2$, we have $\binom{n}{1} = \binom{n+1}{2} - \binom{n}{2}$. This is not the sum we want.

Let's use the identity $\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r}$.

2018 $\binom{n}{2} = \binom{n+1}{3} - \binom{n}{1}$ is not helpful. Let's use $\binom{n}{2} = \binom{n+1}{3} - \binom{n-1}{2}$ is not correct.

Actually, the identity from (i) is $\binom{n}{r-1} = \binom{n+1}{r} - \binom{n}{r}$. For $r = 3$, we have $\binom{n}{2} = \binom{n+1}{3} - \binom{n}{3}$.

$$\begin{aligned}
\text{Sum } S &= \binom{3}{2} + \binom{4}{2} + \binom{5}{2} + \dots + \binom{100}{2} \\
&= \left[\binom{4}{3} - \binom{3}{3} \right] + \left[\binom{5}{3} - \binom{4}{3} \right] + \binom{6}{3} - \binom{5}{3} + \dots + \left[\binom{101}{3} - \binom{100}{3} \right] \\
&= \binom{101}{3} - \binom{3}{3} \\
&= \binom{101}{3} - 1 \\
&= \frac{101 \times 100 \times 99}{3 \times 2 \times 1} - 1 \\
&= 166650 - 1 \\
&= 166649
\end{aligned}$$

Question 5:

i) We simplify the expression $x^n(1+x)^n(1+\frac{1}{x})^n$.

$$\begin{aligned}
LHS &= x^n(1+x)^n \left(\frac{x+1}{x}\right)^n \\
&= x^n(1+x)^n \frac{(1+x)^n}{x^n} \\
&= (1+x)^{2n}
\end{aligned}$$

ii) We expand both sides of $x^n(1+x)^n(1+\frac{1}{x})^n = (1+x)^{2n}$.

$$\begin{aligned}
\text{LHS: } x^n \left(\sum_{r=0}^n \binom{n}{r} x^r \right) \times \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{x}\right)^k \\
&= x^n \left(\sum_{r=0}^n \binom{n}{r} x^r \right) \times \sum_{k=0}^n \binom{n}{k} x^{-k} \\
&= \sum_{r=0}^n \sum_{k=0}^n \binom{n}{r} \binom{n}{k} x^{n+r-k}
\end{aligned}$$

To find the term independent of x (the constant term), we set $n+r-k=0$, so $k=n+r$. Since $k \leq n$ and $r \geq 0$, this only happens when $r=0$ and $k=n$.

Wait, let's look at the product $(1+x)^n(1+\frac{1}{x})^n$. The constant term of $(1+x)^n$ is $\binom{n}{0}$, and the constant term of $(1+x)^n(1+\frac{1}{x})^n$ is $\sum_{r=0}^n \binom{n}{r} \binom{n}{r} = \sum \binom{n}{r}^2$.

Let's re-evaluate the constant term of $x^n(1+x)^n(1+\frac{1}{x})^n$. We want the coefficient of x^{-n} in $(1+x)^n(1+\frac{1}{x})^n$.

$$\begin{aligned}
(1+x)^n(1+\frac{1}{x})^n &= \sum_{r=0}^n \binom{n}{r} x^r \times \sum_{k=0}^n \binom{n}{k} x^{-k} \\
\text{Constant term} &= \sum_{r=0}^n \binom{n}{r} \binom{n}{r} \\
&= \sum_{r=0}^n \binom{n}{r}^2
\end{aligned}$$

Since $x^n \times (\text{constant term of } (1+x)^n(1+\frac{1}{x})^n) = x^n \sum \binom{n}{r}^2$, this is the coefficient of x^n in $(1+x)^{2n}$.

$$\text{Coefficient of } x^n \text{ in } (1+x)^{2n} = \binom{2n}{n}$$

Equating the two:

$$\sum_{r=0}^n \binom{n}{r}^2 = \binom{2n}{n}$$

Which is $1 + ({}^nC_1)^2 + ({}^nC_2)^2 + \dots + ({}^nC_n)^2 = {}^{2n}C_n$.

Question 6:

We use the identity $\cos 2x = \binom{2}{0} \cos^2 x - \binom{2}{1} \sin^2 x$ is not correct. We use $\cos 2x = \cos^2 x - \sin^2 x$.

$$\cos^n 2x = (\cos^2 x - \sin^2 x)^n$$

Using the binomial expansion $(a - b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} (-b)^r$

Let $a = \cos^2 x$

Let $b = \sin^2 x$

$$\begin{aligned} \cos^n 2x &= \sum_{r=0}^n \binom{n}{r} (\cos^2 x)^{n-r} (-\sin^2 x)^r \\ &= \sum_{r=0}^n \binom{n}{r} (-1)^r \cos^{2n-2r} x \sin^{2r} x \\ &= \binom{n}{0} \cos^{2n} x - \binom{n}{1} \cos^{2n-2} x \sin^2 x + \binom{n}{2} \cos^{2n-4} x \sin^4 x - \dots + (-1)^n \binom{n}{n} \sin^{2n} x \end{aligned}$$

Question 7:

i) We substitute $x = 1$ into $(1 + x)^n = \sum_{r=0}^n \binom{n}{r} x^r$.

$$\begin{aligned} (1 + 1)^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \\ 2^n &= \binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{n} \end{aligned}$$

ii) We substitute $x = -1$ into $(1 + x)^n = \sum_{r=0}^n \binom{n}{r} x^r$.

$$\begin{aligned} (1 - 1)^n &= \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots \\ 0 &= \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots \end{aligned}$$

$$\text{Rearrange the terms: } \binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots$$

Question 8:

a) We simplify the expression $(1 - x)^5 (1 + \frac{1}{x})^5$.

$$\begin{aligned}
(1-x)^5 \left(1 + \frac{1}{x}\right)^5 &= (1-x)^5 \left(\frac{x+1}{x}\right)^5 \\
&= (1-x)^5 \frac{(x+1)^5}{x^5} \\
&= \frac{((1-x)(1+x))^5}{x^5} \\
&= \frac{(1-x^2)^5}{x^5} \\
&= \left(\frac{1-x^2}{x}\right)^5 \\
&= \left(\frac{1}{x} - x\right)^5
\end{aligned}$$

b) We expand $\left(\frac{1}{x} - x\right)^5$ using the binomial theorem:

$$\begin{aligned}
\left(\frac{1}{x} - x\right)^5 &= \binom{5}{0} \left(\frac{1}{x}\right)^5 (-x)^0 + \binom{5}{1} \left(\frac{1}{x}\right)^4 (-x)^1 + \binom{5}{2} \left(\frac{1}{x}\right)^3 (-x)^2 + \binom{5}{3} \left(\frac{1}{x}\right)^2 (-x)^3 \\
&\quad + \binom{5}{4} \left(\frac{1}{x}\right)^1 (-x)^4 + \binom{5}{5} \left(\frac{1}{x}\right)^0 (-x)^5 \\
&= \binom{5}{0} x^{-5} - \binom{5}{1} x^{-3} + \binom{5}{2} x^{-1} - \binom{5}{3} x^1 + \binom{5}{4} x^3 - \binom{5}{5} x^5
\end{aligned}$$

c) The coefficient of x^3 in $\left(\frac{1}{x} - x\right)^5$ is $\binom{5}{4}$.

Now, we find the coefficient of x^3 in $(1-x)^5 \left(1 + \frac{1}{x}\right)^5$ by expanding both:

$$\begin{aligned}
(1-x)^5 &= \sum_{r=0}^5 \binom{5}{r} (-x)^r \\
\left(1 + \frac{1}{x}\right)^5 &= \sum_{k=0}^5 \binom{5}{k} x^{-k}
\end{aligned}$$

To get x^3 , we need $r - k = 3$, so $r = k + 3$.

$$\begin{aligned}
\text{Coeff of } x^3 &= \sum_{k=0}^2 \binom{5}{k+3} (-1)^{k+3} \binom{5}{k} \\
&= \binom{5}{3} (-1)^3 \binom{5}{0} - \binom{5}{4} (-1)^4 \binom{5}{1} + \binom{5}{5} (-1)^5 \binom{5}{2} \\
&= -\binom{5}{3} \binom{5}{0} - \binom{5}{4} \binom{5}{1} - \binom{5}{5} \binom{5}{2}
\end{aligned}$$

Wait, let's check the signs. The expansion of $(1-x)^5$ is $\binom{5}{0} - \binom{5}{1}x + \binom{5}{2}x^2 - \binom{5}{3}x^3 + \binom{5}{4}x^4 - \binom{5}{5}x^5$.

$$\text{Coeff of } x^3 = \binom{5}{4} \binom{5}{1} - \binom{5}{5} \binom{5}{2} - \binom{5}{3} \binom{5}{0} \text{ (after checking indices and signs)}$$

Since the coefficient of x^3 is $\binom{5}{4}$, we have:

$$\binom{5}{4}\binom{5}{1} - \binom{5}{5}\binom{5}{2} - \binom{5}{3}\binom{5}{0} = \binom{5}{4}$$

Question 9:

i) We simplify the expression $(1+x)^m(1-\frac{1}{x})^m$.

$$\begin{aligned} (1+x)^m(1-\frac{1}{x})^m &= (1+x)^m(\frac{x-1}{x})^m \\ &= \frac{(1+x)^m(x-1)^m}{x^m} \\ &= \frac{(x^2-1)^m}{x^m} \\ &= (\frac{x^2-1}{x})^m \\ &= (x-\frac{1}{x})^m \end{aligned}$$

ii) We find the term independent of x in $(x-\frac{1}{x})^m$ where $m = 2002$.

$$\begin{aligned} \text{General term of } (x-\frac{1}{x})^{2002} &= \binom{2002}{k} x^{2002-k} (-x^{-1})^k \\ &= \binom{2002}{k} (-1)^k x^{2002-2k} \end{aligned}$$

For the term to be independent of x , we set $2002 - 2k = 0$, so $k = 1001$.

$$\begin{aligned} \text{Constant term} &= \binom{2002}{1001} (-1)^{1001} \\ &= -\binom{2002}{1001} \end{aligned}$$

Now, we find the constant term in $(1+x)^{2002}(1-\frac{1}{x})^{2002}$.

$$\begin{aligned} \text{Constant term} &= \sum_{k=0}^{2002} \binom{2002}{k} x^k \times \binom{2002}{k} (-x^{-1})^k \\ &= \sum_{k=0}^{2002} \binom{2002}{k} \binom{2002}{k} (-1)^k \\ &= \sum_{k=0}^{2002} (-1)^k \binom{2002}{k}^2 \end{aligned}$$

Equating the two constant terms:

$$\binom{2002}{0}^2 - \binom{2002}{1}^2 + \binom{2002}{2}^2 - \dots + \binom{2002}{2002}^2 = -\binom{2002}{1001}$$

iii) We use the symmetry $\binom{n}{k} = \binom{n}{n-k}$.

$$\begin{aligned} \text{Sum } S_{total} &= \sum_{k=0}^{2002} (-1)^k \binom{2002}{k}^2 \\ &= \left[\binom{2002}{0}^2 - \binom{2002}{1}^2 + \dots - \binom{2002}{1000}^2 \right] + (-1)^{1001} \binom{2002}{1001}^2 \\ &\quad + \left[\binom{2002}{1001}^2 - \binom{2002}{1002}^2 + \dots + \binom{2002}{2002}^2 \right] \end{aligned}$$

But the sum in the question is from $k = 0$ to 1001.

$$\text{Let } S_{total} = \sum_{k=0}^{2002} (-1)^k \binom{2002}{k}^2 = -\binom{2002}{1001}.$$

$$\text{Since } (-1)^k \binom{2002}{k}^2 = (-1)^{2002-k} \binom{2002}{2002-k}^2$$

$$\begin{aligned} \text{Total sum } S_{total} &= 2 \sum_{k=0}^{1000} (-1)^k \binom{2002}{k}^2 + (-1)^{1001} \binom{2002}{1001}^2 \\ &\quad - \binom{2002}{1001} = 2 \sum_{k=0}^{1000} (-1)^k \binom{2002}{k}^2 - \binom{2002}{1001}^2 \\ 2 \sum_{k=0}^{1000} (-1)^k \binom{2002}{k}^2 &= \binom{2002}{1001}^2 - \binom{2002}{1001} \end{aligned}$$

$$\begin{aligned} \text{Sum from 0 to 1001} &= \sum_{k=0}^{1000} (-1)^k \binom{2002}{k}^2 + (-1)^{1001} \binom{2002}{1001}^2 \\ &= \frac{1}{2} \left[\binom{2002}{1001}^2 - \binom{2002}{1001} \right] - \binom{2002}{1001}^2 \\ &= \frac{1}{2} \binom{2002}{1001}^2 - \frac{1}{2} \binom{2002}{1001} - \binom{2002}{1001}^2 \\ &= -\frac{1}{2} \binom{2002}{1001}^2 - \frac{1}{2} \binom{2002}{1001} \\ &= -\frac{1}{2} \binom{2002}{1001} \left[\binom{2002}{1001} + 1 \right] \end{aligned}$$